IEEE VTS Motor Vehicles Challenge 2021

Energy Management of A Dual-Motor All-Wheel Drive Electric Vehicle

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Content

1. Introduction
2. Dual-motor AWD EV: modeling and control
3. Energy management strategy: design and scoring
4. Conclusion

Introduction

to a Challenge that you may have fun

IEEE VTS Motor Vehicles Challenge 2021

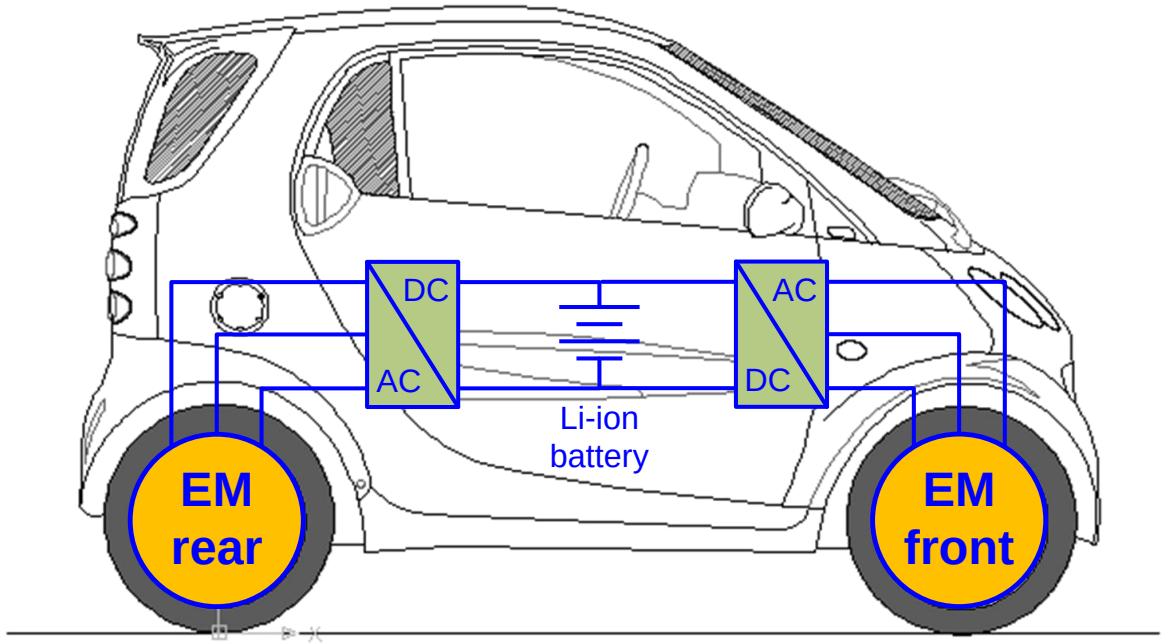
Vehicle model reference

Urban electric vehicle VUE



Studied configuration

Dual-motor all-wheel drive (AWD) electric vehicle (EV)



Objective of the Challenge:

An energy management strategy (EMS) that saves the most battery state-of-charge (SoC)

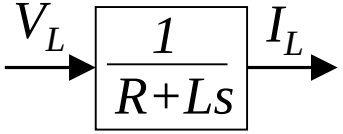
Tool for modeling and control: Energetic Macroscopic Representation (EMR)



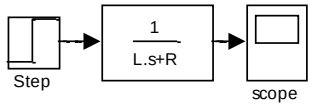
System

$$v_l = L\frac{d}{dt}i_L + Ri_L$$

Mathematical model

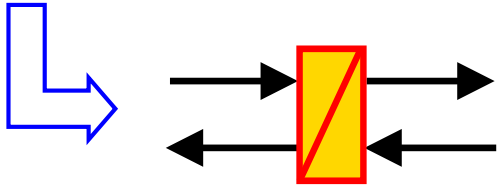


Model representation

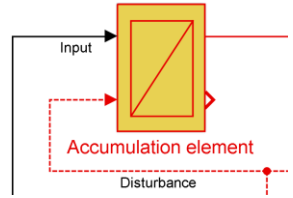


Implementation (Simulation)

EMR: a graphical formalism for **model representation**



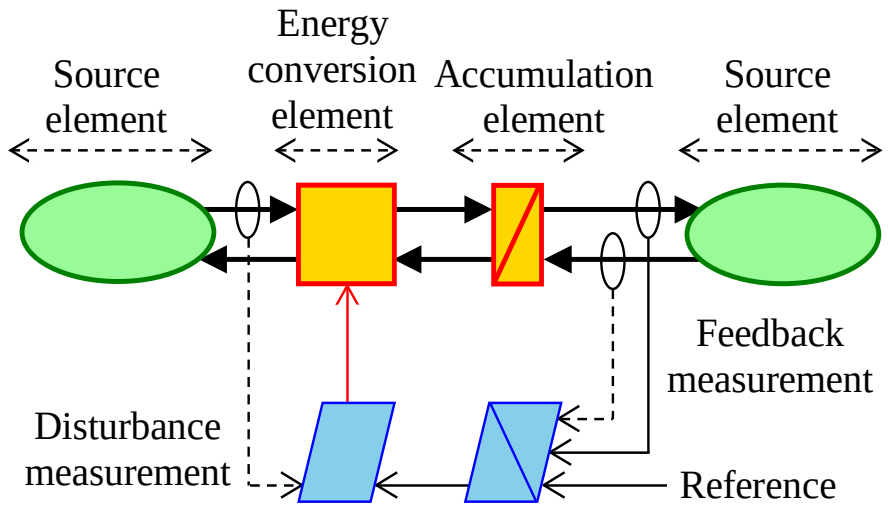
Model representation



Implementation (Simulink®-based EMR library)

not only **model**

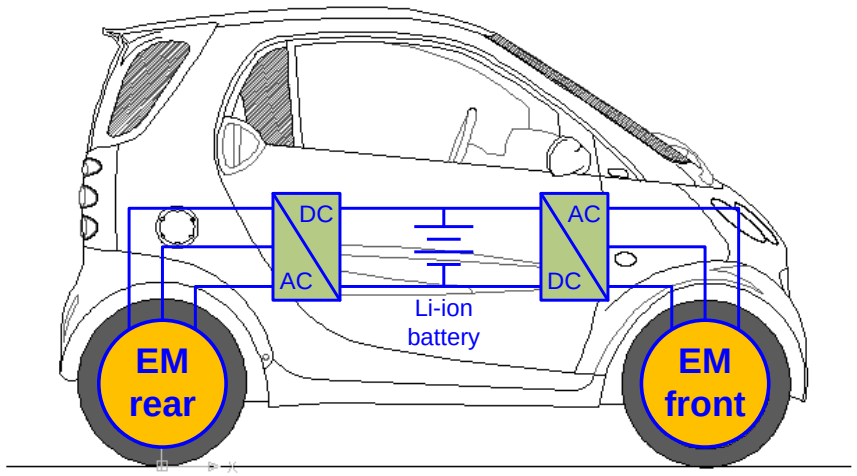
but also **control**



What we provide:

Modeling and control of the studied vehicle

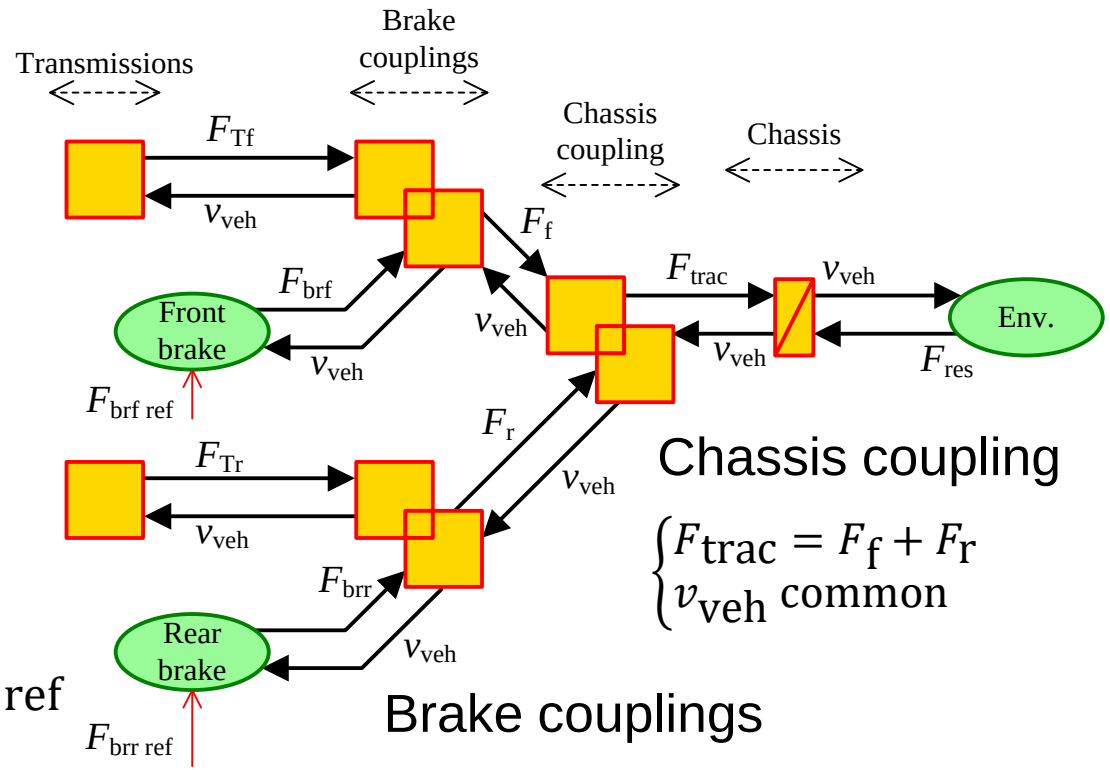
EMR-based modeling and control of the vehicle: Chassis and Drivetrain



Drivetrain (transmissions)

$$\begin{cases} F_{\mathrm{Tx}} = \Gamma_{\mathrm{EDx}} \eta_{\mathrm{tran}}^{k\mathrm{trx}} k_{\mathrm{tran\ x}} \\ \Omega_{\mathrm{EDx}} = v_{\mathrm{veh}} k_{\mathrm{tran\ x}} \end{cases}$$

$$\text{with } k_{\mathrm{trx}} = \begin{cases} 1 \text{ if } F_{\mathrm{Tx}} v_{\mathrm{veh}} \geq 0 \\ -1 \text{ if } F_{\mathrm{Tx}} v_{\mathrm{veh}} < 0 \end{cases}$$



Brakes

$$F_{\mathrm{brx}} = F_{\mathrm{brx\ ref}}$$

Chassis coupling

$$\begin{cases} F_{\mathrm{trac}} = F_{\mathrm{f}} + F_{\mathrm{r}} \\ v_{\mathrm{veh}} \text{ common} \end{cases}$$

Brake couplings

$$\begin{cases} F_{\mathrm{x}} = F_{\mathrm{Tx}} + F_{\mathrm{brx}} \\ v_{\mathrm{veh}} \text{ common} \end{cases}$$

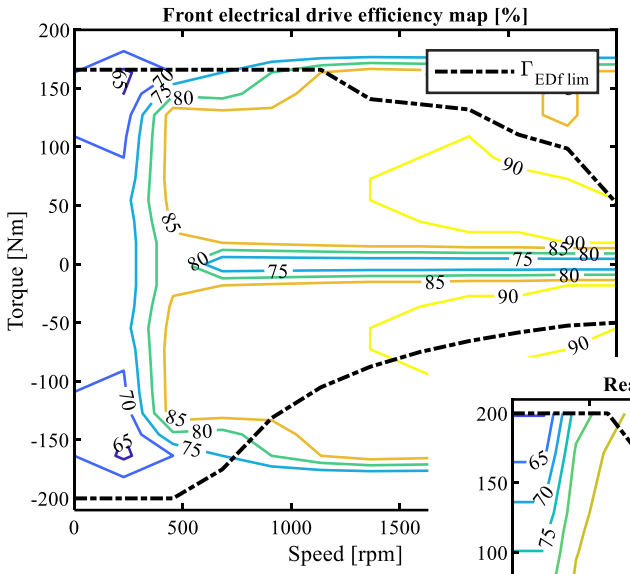
Chassis

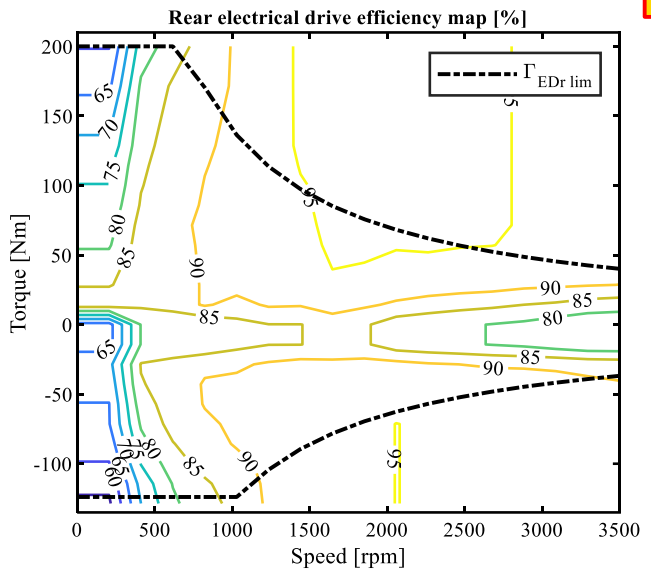
$$v_{\mathrm{veh}} = \frac{1}{M_{\mathrm{veh}}} \int_0^t (F_{\mathrm{trac}} - F_{\mathrm{res}}) dt$$

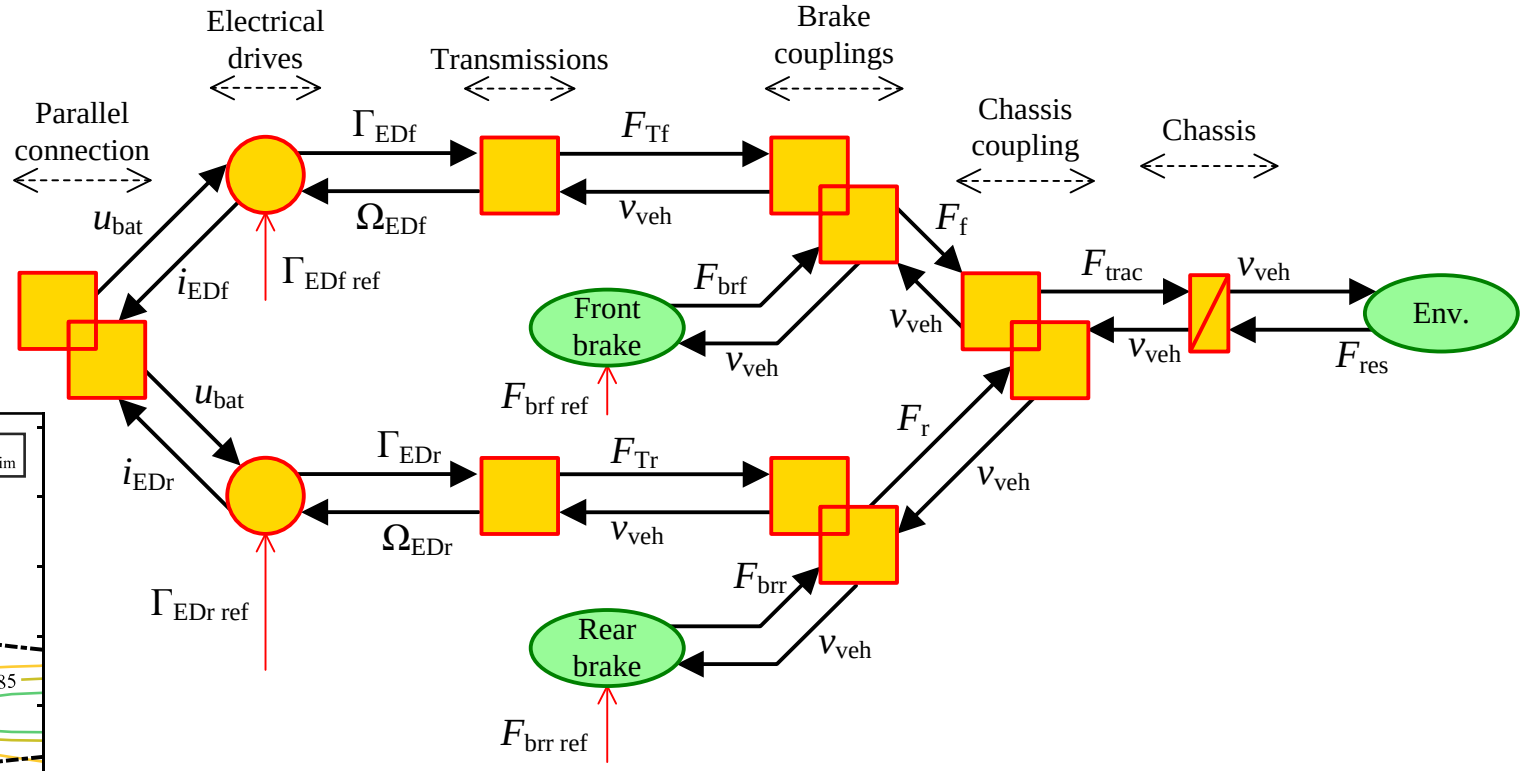
Environment

$$F_{\mathrm{res}} = k_{\mathrm{roll}} M_{\mathrm{veh}} g + 0.5 \rho c_{\mathrm{x}} A \left(v_{\mathrm{veh}} + v_{\mathrm{wind}}\right)^2 + M_{\mathrm{veh}} g \sin \alpha$$

EMR-based modeling and control of the vehicle: Electrical drives







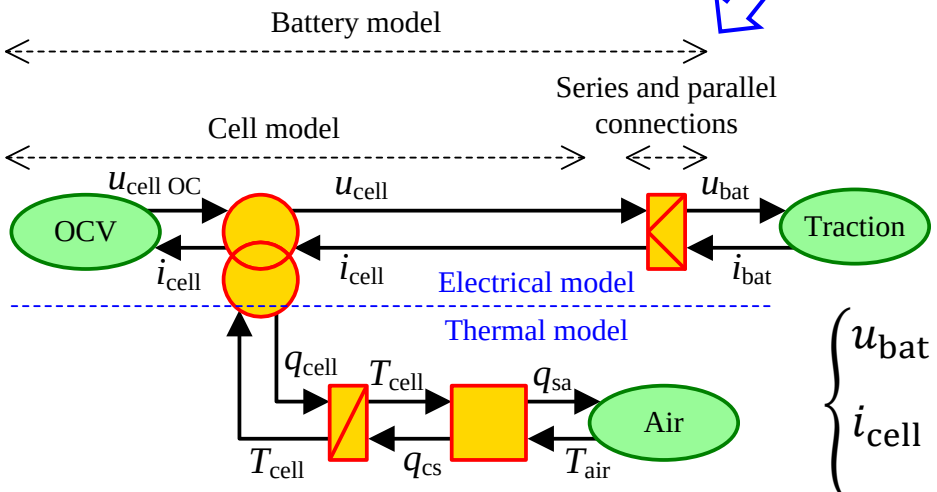
$$\begin{cases} \Gamma_{EDx} = \Gamma_{EDx\,ref} \\ i_{EDx} = \dfrac{\Gamma_{EDx}\Omega_{EDx}}{u_{bat}\eta_{EDx}^{k_{EDx}}} \text{ with } k_{EDx} = \begin{cases} 1 \text{ if } \Gamma_{EDx}\Omega_{EDx} \geq 0 \\ -1 \text{ if } \Gamma_{EDx}\Omega_{EDx} < 0 \end{cases} \end{cases}$$

Front and rear electrical drives:

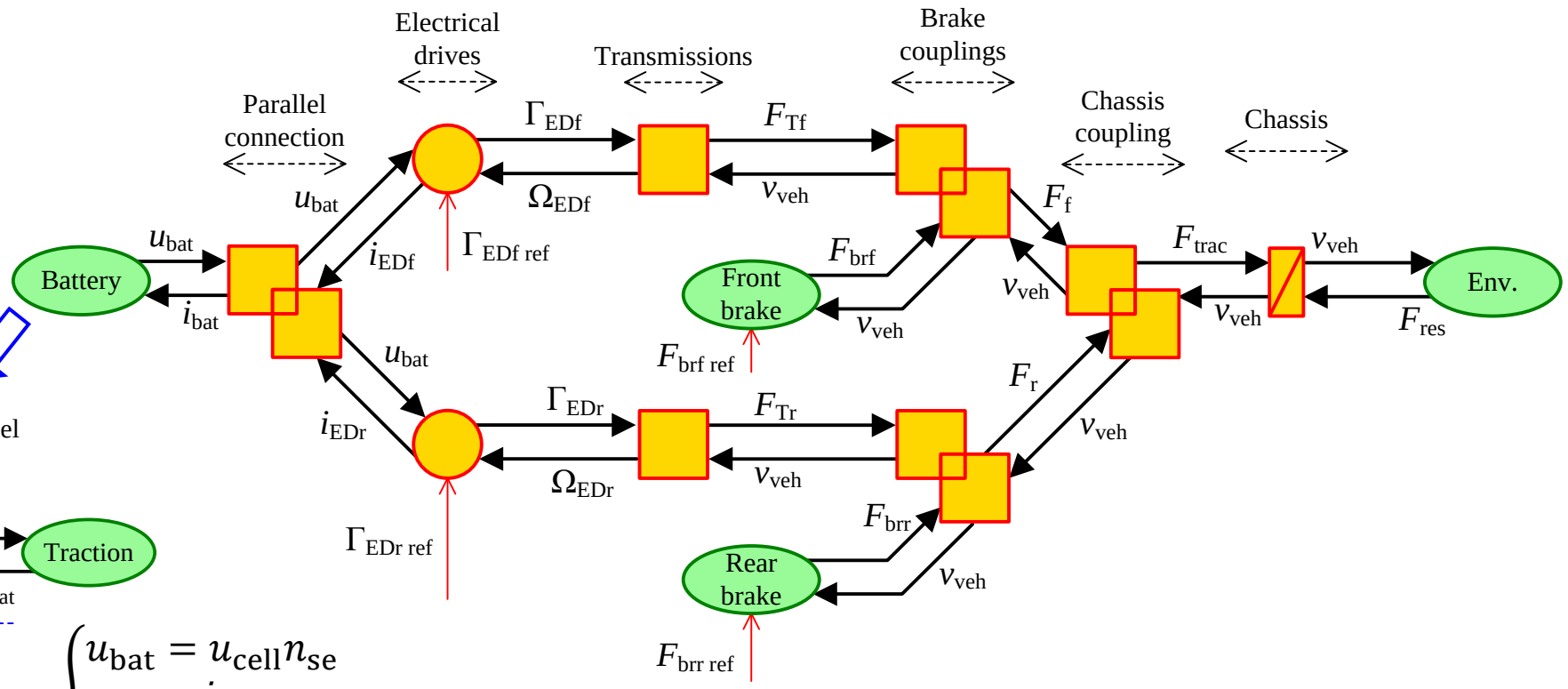
- Static model given by efficiency maps
- Note the different shapes of power envelopes

EMR-based modeling and control of the vehicle: Battery

$$\begin{cases} u_{\text{cell}} = u_{\text{cell OC}}(SoC_{\text{cell}}, T_{\text{cell}}) \\ \qquad -r_{\text{cell}}(SoC_{\text{cell}}, T_{\text{cell}})\, i_{\text{cell}} \\ SoC_{\text{cell}} = SoC_{\text{cell}}(0) - \dfrac{1}{C_{\text{eq}}} \displaystyle\int_0^t i_{\text{cell}}\, dt \end{cases}$$



$$\begin{cases} u_{\text{bat}} = u_{\text{cell}} n_{\text{se}} \\ i_{\text{cell}} = \dfrac{i_{\text{bat}}}{n_{\text{pa}}} \end{cases}$$



$$q_{\text{cell}} = \frac{r_{\text{cell}}(SoC_{\text{cell}}, T_{\text{cell}})\, i_{\text{cell}}^2}{T_{\text{cell}}}$$

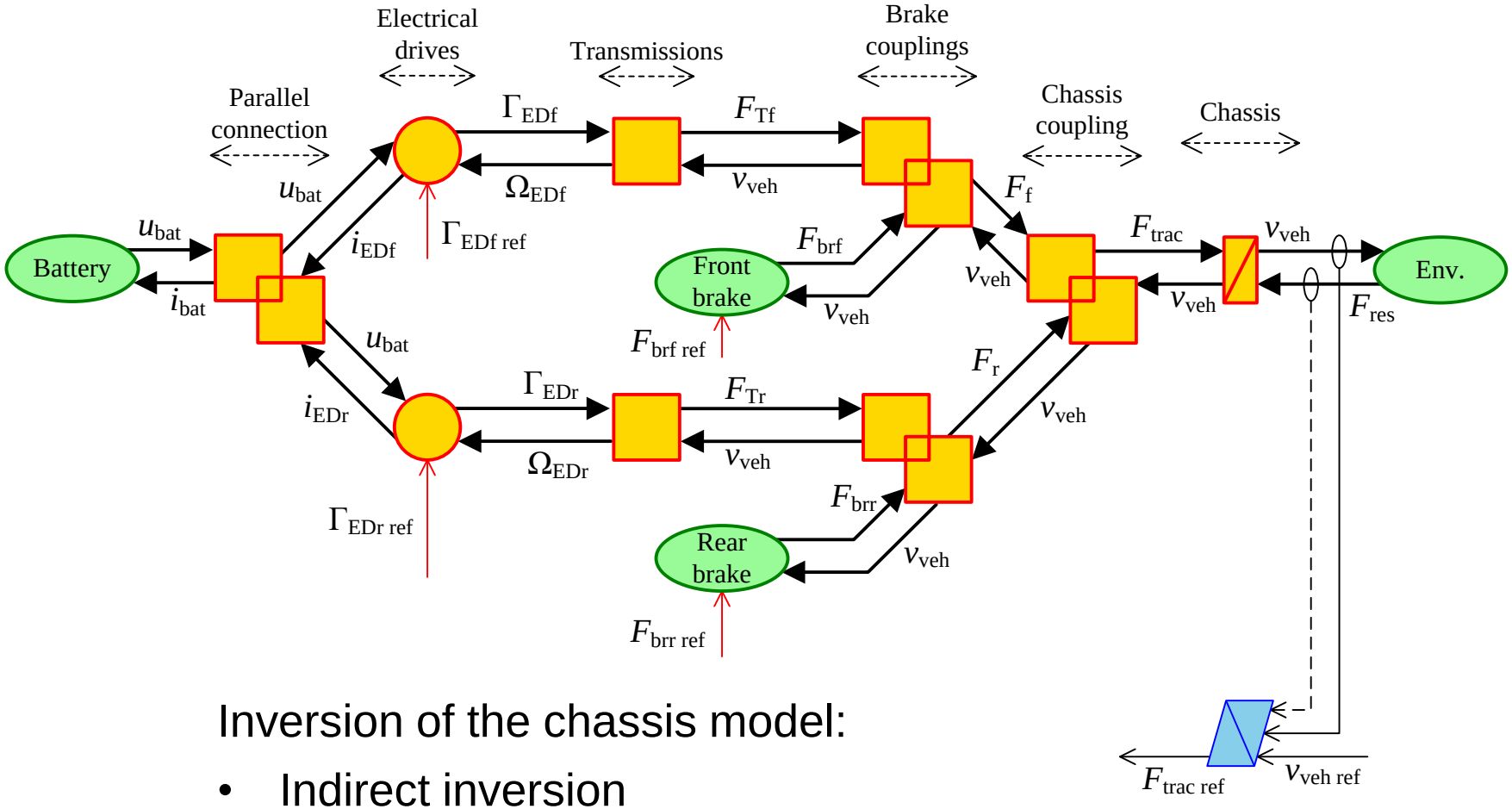
$$T_{\text{cell}} = T_{\text{cell}}(0)\, e^{\frac{1}{C_{\text{th}}} \int_0^t (q_{\text{cell}} - q_{\text{cs}})\, dt}$$

$$\begin{cases} q_{\text{cs}} = \dfrac{T_{\text{cell}} - T_{\text{air}}}{r_{\text{th}} T_{\text{cell}}} \\ q_{\text{sa}} = \dfrac{T_{\text{cell}} - T_{\text{air}}}{r_{\text{th}} T_{\text{air}}} \end{cases}$$

Battery electro-thermal model:

- Nonlinear characteristics given by look-up tables
- Coupling between electrical and thermal models

EMR-based modeling and control of the vehicle: Speed control

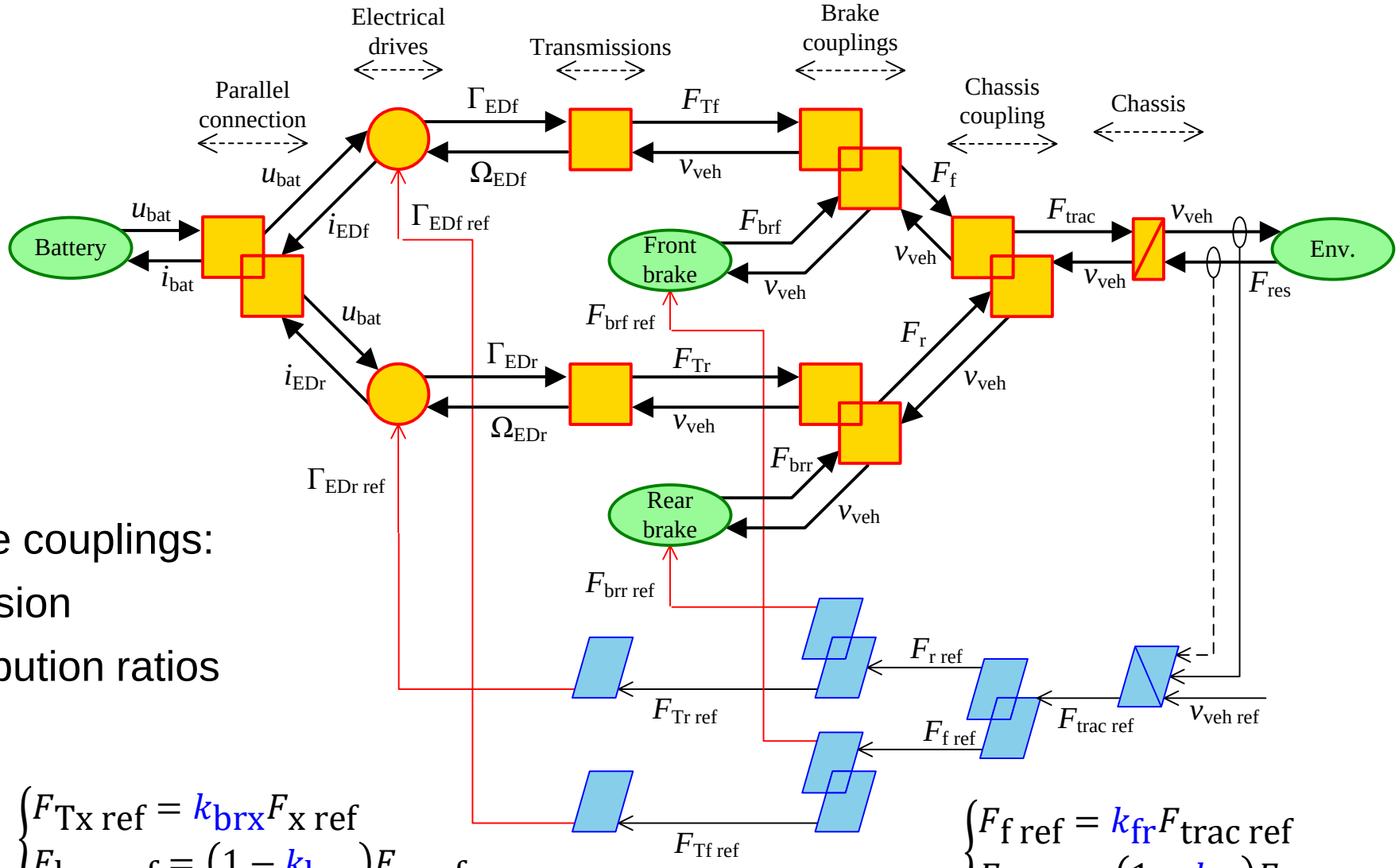


Inversion of the chassis model:

- Indirect inversion
- Realized by a closed-loop controller

$$F_{\text{trac ref}} = F_{\text{res mea}} + k_{\text{I}} \int_0^t (v_{\text{veh ref}} - v_{\text{veh mea}})dt - k_{\text{P}} v_{\text{veh mea}}$$

EMR-based modeling and control of the vehicle: Drivetrain control



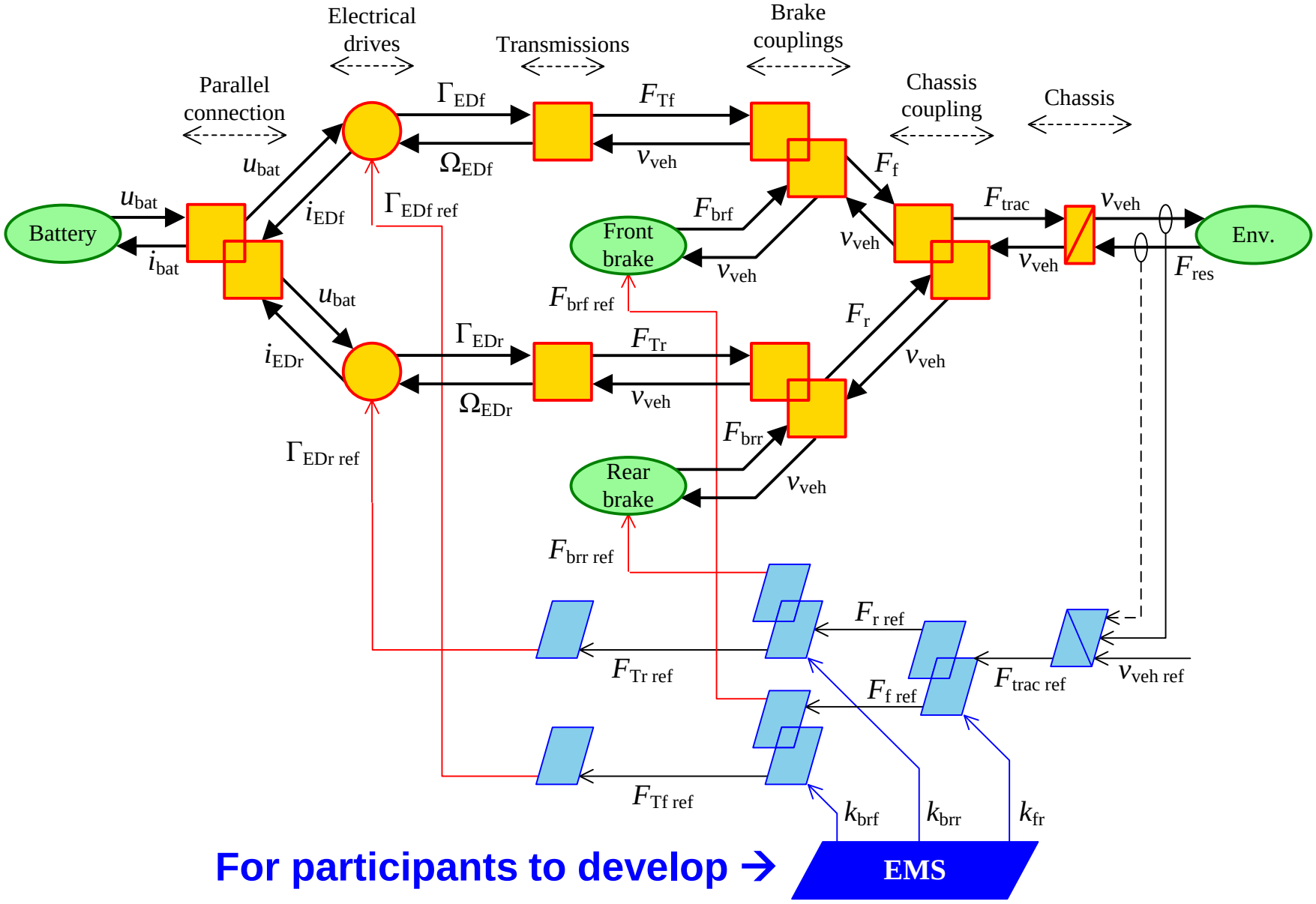
Inversion of the couplings:

- Direct inversion
- Force distribution ratios

$$\begin{cases} F_{\mathrm{Tx\ ref}} = k_{\mathrm{brx}} F_{\mathrm{x\ ref}} \\ F_{\mathrm{brx\ ref}} = (1 - k_{\mathrm{brx}}) F_{\mathrm{x\ ref}} \end{cases}$$

$$\begin{cases} F_{\mathrm{f\ ref}} = k_{\mathrm{fr}} F_{\mathrm{trac\ ref}} \\ F_{\mathrm{r\ ref}} = (1 - k_{\mathrm{fr}}) F_{\mathrm{trac\ ref}} \end{cases}$$

EMR-based modeling and control of the vehicle: Strategy

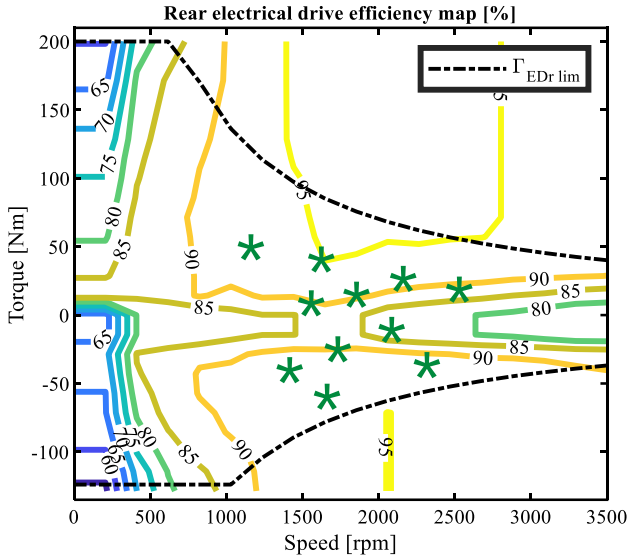


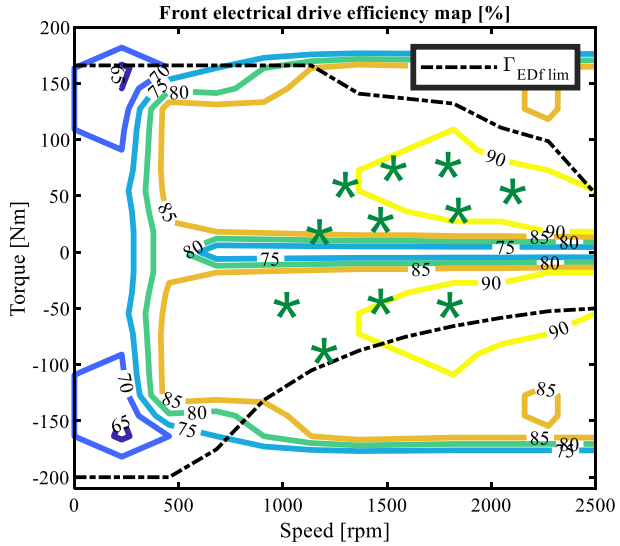
For participants to develop →

The energy management strategy

What you should do
How we score

Energy management problem formulation and Strategy evaluation





$$[k_{\mathrm{fr}} \quad k_{\mathrm{brf}} \quad k_{\mathrm{brr}}] = \arg\min\left[SoC_{\mathrm{bat}}(0) - SoC_{\mathrm{bat}}(t_{\mathrm{f}})\right]$$

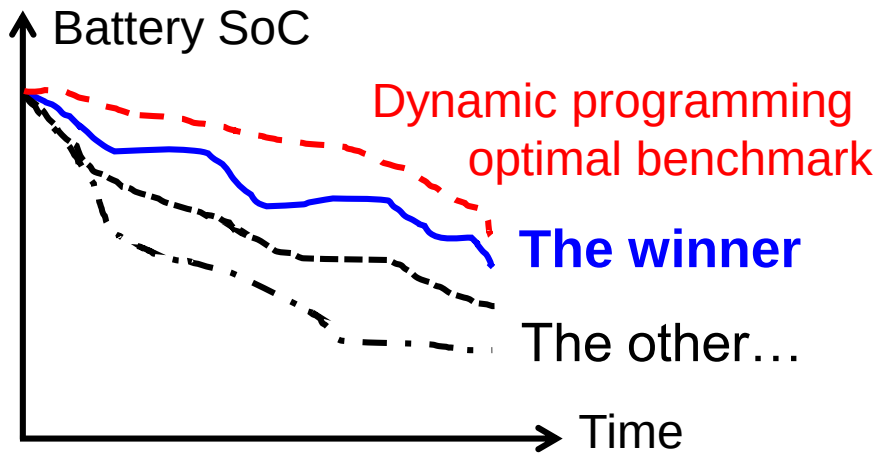
$$\text{s. t.} \quad \Gamma_{\mathrm{EDf\,min}} \leq \Gamma_{\mathrm{EDf}} \leq \Gamma_{\mathrm{EDf\,max}}$$

$$\Gamma_{\mathrm{EDr\,min}} \leq \Gamma_{\mathrm{EDf}} \leq \Gamma_{\mathrm{EDr\,max}}$$

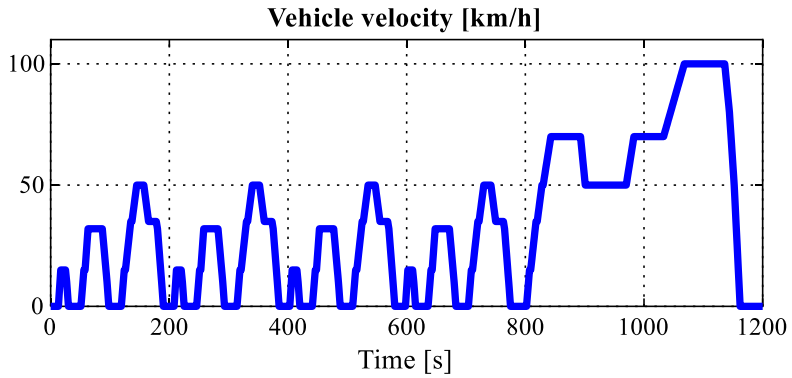
$$T_{\mathrm{cell\,min}} \leq T_{\mathrm{cell}} \leq T_{\mathrm{cell\,max}}$$

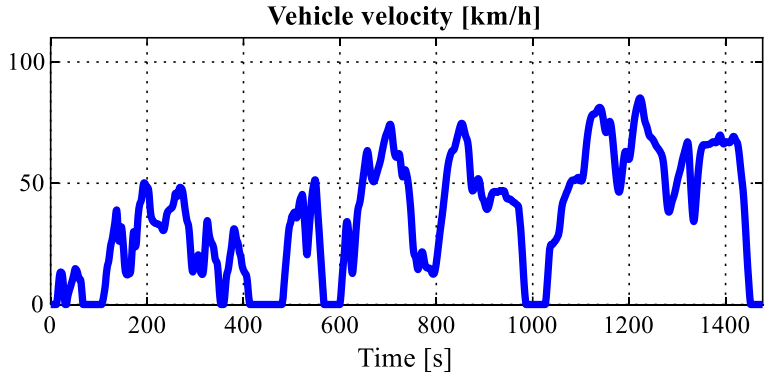
The ultimate objective:

Minimize the battery SoC degradation



Given samples of driving cycles





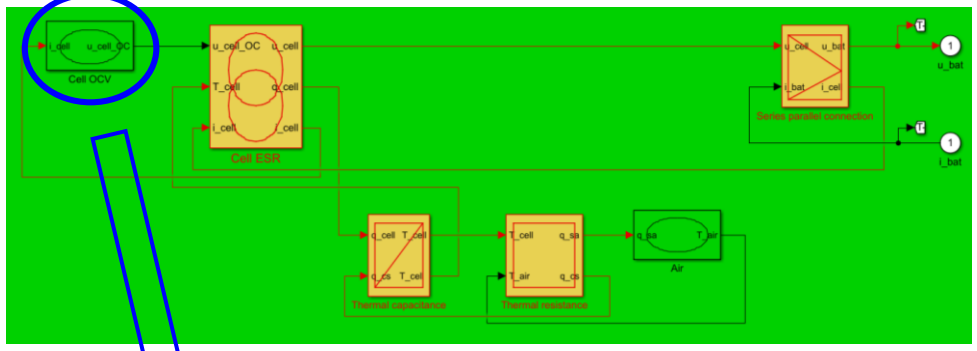
Scoring cycle ?

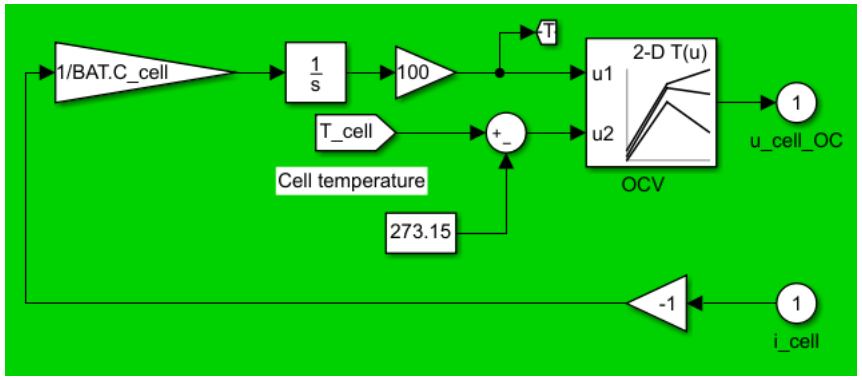
→ unknown

MATLAB/Simulink-based simulation common framework with EMR library

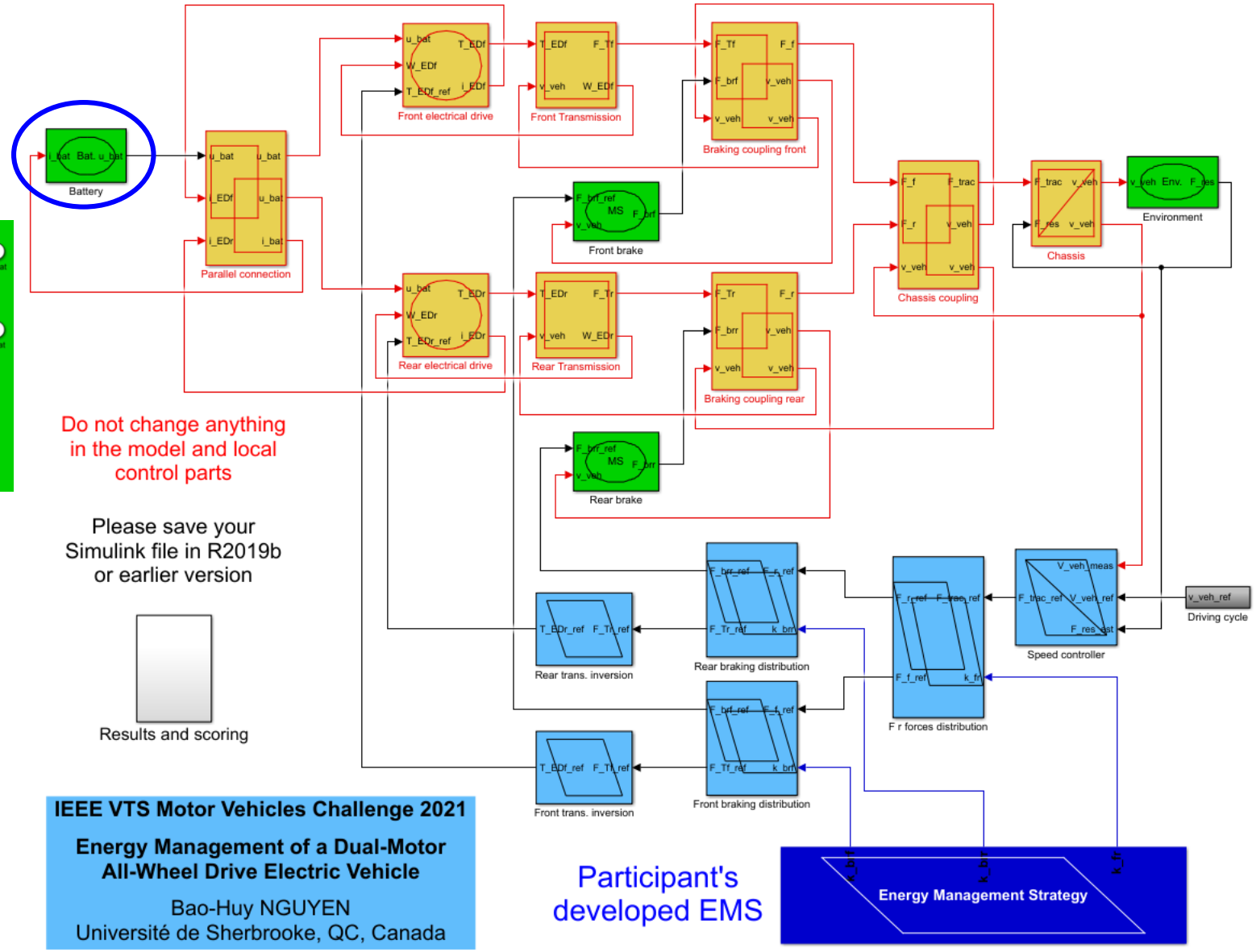
Free download !

Battery electro-thermal model





Example of model implementation in EMR blocks



What you should keep in mind

To conclude

The Challenge:

- Studied vehicle: a dual-motor all-wheel drive EV
- Objective: minimize the battery SoC degradation

Awards:

- All can participate (students, academics, industry)
- Only IEEE VTS members can receive awards:
  - First prize: Up to US$3500
  - Second prize: Up to US$1500

  to participate IEEE VPPC 2021

→ **Join us !**

https://vtsociety.org/

Important dates:

- Registration: December 20, 2020
- Submission: February 22, 2021
- Results: March 15, 2021

→ **You should start right now !**

Go to http://www.motorvehicleschallenge.org/

Contact us: vtschallenge2021@usherbrooke.ca

Thank you
Have fun
Join us
and see you in VPPC 2021 !



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